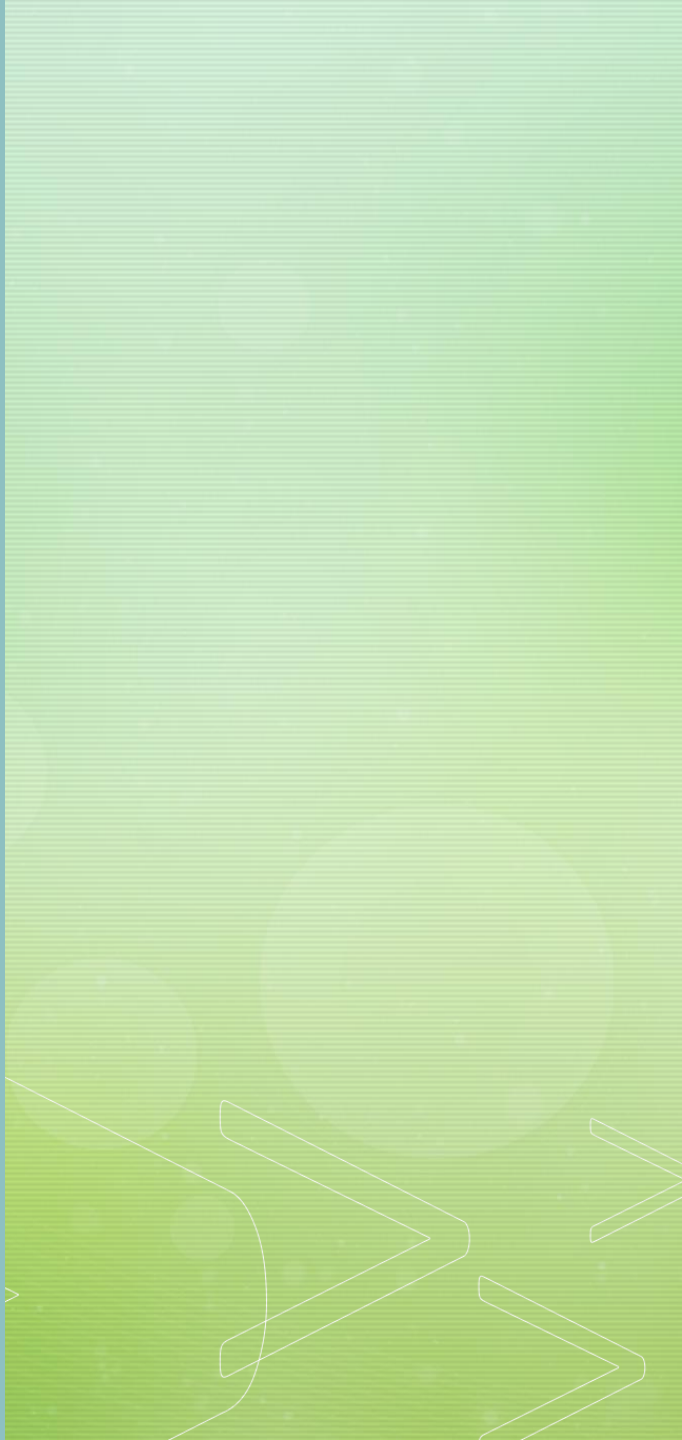




Unit 01: INTRODUCTION TO

Scientific Research



COURSE CONTENTS



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WHAT IS RESEARCH?



“The systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions.” [Oxford English Dictionary](#)

WHAT IS RESEARCH?

- A more **academic** interpretation is that research involves building upon the works of those scholars who came before us to **advance the frontiers of knowledge**.
- One's research should not just return the work of previous scholars.

Nicholas Walliman, *Research Methods: The Basics*
(2010)

WHAT YOU CAN DO WITH RESEARCH

- **Describe.** Descriptive research relies on observation as a means of collecting data. It attempts to examine situations in order to establish what is the norm, i.e. what can be predicted to happen again under the same circumstances.
- **Explain.** This is a descriptive type of research specifically designed to deal with complex issues. It aims to move beyond 'just getting the facts' in order to make sense of the myriad other elements involved, such as human, political, social, cultural and contextual.
- **Correlate.** The relationships between two phenomena are investigated to see whether and how they influence each other. The relationship might be just a loose link at one extreme or a direct link when one phenomenon causes another. These are measured as levels of association

Nicholas Walliman, *Research Methods: The Basics* (2010)

WHAT YOU CAN DO WITH RESEARCH

- **Predict.** This can sometimes be done in research areas where correlations are already known. **Predictions** of possible future behavior or events are made on the basis that if there has been a strong relationship between two or more characteristics or events in the past, then these should exist in similar circumstances in the future, leading to predictable outcomes.
- **Control.** Once you understand an event or situation, you may be able to find ways to **control** it. For this you need to know what the cause and effect relationships are and that you are capable of exerting control over the vital ingredients. All of technology relies on this ability to control.
- ...

Nicholas Walliman, *Research Methods: The Basics* (2010)

RESEARCH THEORY

- Research is about producing new knowledge to build up a picture of the world.
- Then it is obvious that we should have a view on what knowledge is (**metaphysics**) and how we can acquire knowledge (**epistemology**).
- These views is based on the **philosophical position** that we hold.

WHY PHILOSOPHY?

“... the alternative to philosophy is not no philosophy, but bad philosophy. The ‘unphilosophical’ person has an unconscious philosophy, which they apply in their practice — whether of science or politics or daily life.”

Andrew Collier, *Critical Realism: An Introduction to Roy Bhaskar's Philosophy* (1994)

WHY PHILOSOPHY?

- A philosophical position holds a view about **reality**.
- This view, in turn, determine what is **legitimate knowledge**.
- In order to evaluate a research report, we need to consider the philosophy that underpins it.
- This will help to conceive the implicit underlying assumptions that the research is based on, and thus enable us to judge the methods used and the conclusions reached.

“The branch of philosophy that deals with the first principles of things, including abstract concepts such as being, knowing, substance, cause, identity, time, and space.”

Oxford English Dictionary

- **Idealism** and **Materialism** are two opposite positions which emphasize the importance of **the mind** and **material things**, respectively.



“**Idealism** is the group of metaphysical philosophies which assert that reality, or reality as humans can know it, is fundamentally mental, mentally constructed, or otherwise immaterial.

Epistemologically, idealism manifests as a skepticism about the possibility of knowing any mind-independent thing.” [Wikipedia](#)

“**Materialism** is a form of philosophical monism which holds that matter is the fundamental substance in nature, and that all things, including mental states and consciousness, are results of material interactions.” [Wikipedia](#)

“The theory of knowledge, especially with regard to its methods, validity, and scope. Epistemology is the investigation of what distinguishes justified belief from opinion.”

Oxford English Dictionary

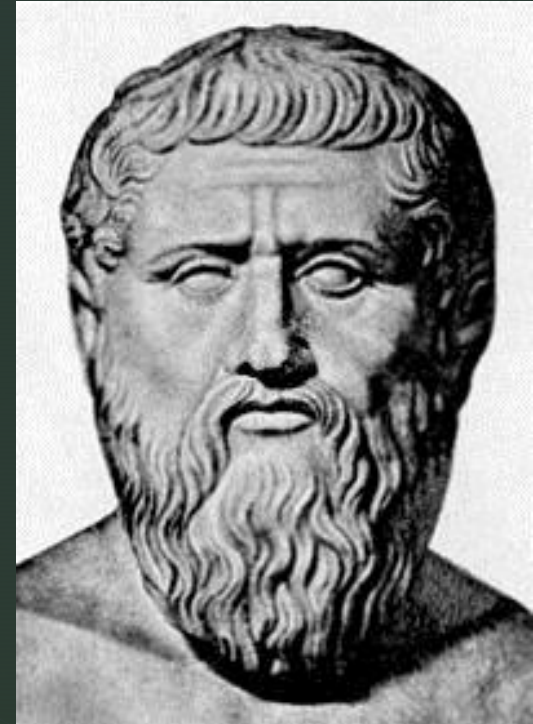
- As the methods of gaining knowledge, **empiricism** and **rationalism** are two basic approaches.

EMPIRICISM & RATIONALISM

- **Empiricism** is “the theory that all knowledge is derived from sense-experience.” [Oxford English Dictionary](#)
 - Using **inductive reasoning** to gain knowledge.
- **Rationalism** is “the theory that reason rather than experience is the foundation of certainty in knowledge.” [Oxford English Dictionary](#)
 - Using **deductive reasoning** to gain knowledge.
- The benefits of these two opposing approaches have been argued since the Ancient Greece.

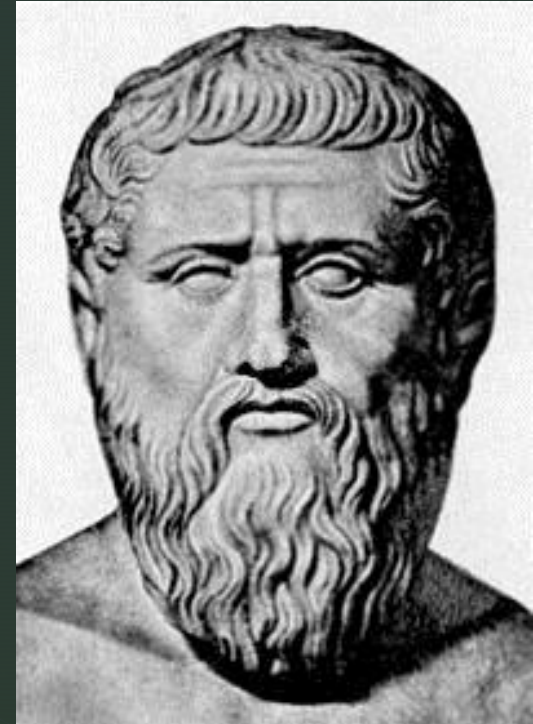
PLATO and RATIONALISM

- **Plato** (Greek: Πλάτων, Plátōn, "wide, broad-browed") (428/427 BC – 348/347 BC), was a Classical Greek philosopher. Together with his teacher, Socrates, and his student, Aristotle, Plato helped to lay the philosophical foundations of Western culture. ([Wikipedia](#))
- “Another view, generally associated with **Plato** (*Republic* 479e-484c), locates the superiority of a **priori knowledge** in the objects known. What we know by **reason alone**, a Platonic form, say, is superior in an important metaphysical way, e.g. unchanging, eternal, perfect, a higher degree of being, to what we are aware of through sense experience.” ([Stanford Encyclopedia of Philosophy](#))



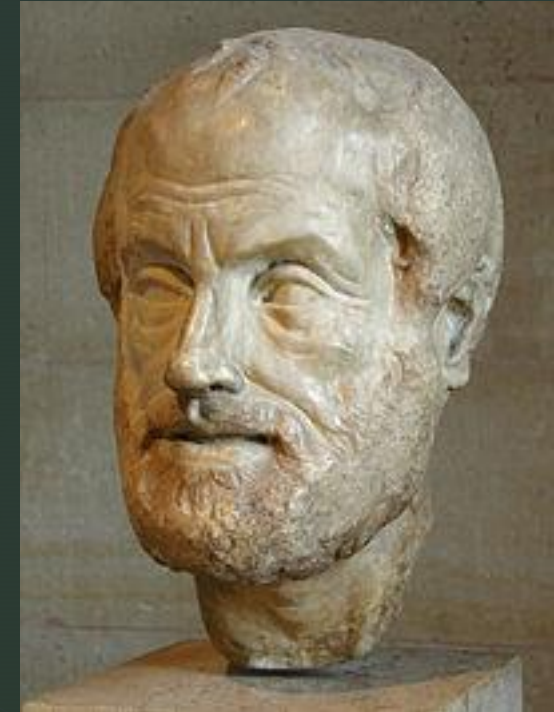
PLATO and RATIONALISM

- Rationalists claim that there are significant ways in which our concepts and knowledge are gained independently of sense experience.
- ***The Intuition/Deduction Thesis:*** Some propositions in a particular subject area, S, are knowable by us by intuition alone; still others are knowable by being deduced from intuited propositions.
- We intuit, for example, that the number three is prime and that it is greater than two. We then deduce from this knowledge that there is a prime number greater than two.



ARISTOTLE and EMPIRICISM

- **Aristotle** (Greek: Ἀριστοτέλης Aristotélēs) (384 BC – 322 BC) was a Greek philosopher, a student of Plato and teacher of Alexander the Great. ([Wikipedia](#))
- Aristotle believes that knowledge comes from experience.
- “For Aristotle, the Platonic, universal ideal is to be found in particular things, what he calls the **essence** of things. Using the concept of essence, Aristotle **reconciles abstract thought with observation**. In Aristotelian science, we find the beginnings of a primitive **inductive method**, although one that is based on collections of objects rather than experimentation.”
(<https://www.psychegames.com>)



INDUCTIVE REASONING

- **Inductive reasoning** starts from specific observations or **sensory experiences** and then develops a **general conclusion** from them.
- **Francis Bacon** (1561-1626) stated that one should consult nature, and not rely on the writings of ancient philosophers such as Aristotle or on the Bible.
- The **scientific revolution** in the seventeenth century was based on this approach, led by such scientists as Galileo and Newton.
- Mendel's discovery of genetics and Darwin's theory of evolution are perhaps the most famous generalizations in the form of **theories** that are, even by them, claimed to be developed through inductive reasoning.

All the giraffes that I have
seen have very long necks.
(Repeated observations)

Therefore I conclude that all
giraffes have long necks.
(Conclusion)

PROBLEMS of INDUCTION

- How many observations must be made before we can reasonably draw a conclusion that is reliable enough to generalize from.
- How many situations and under which conditions should the observations be made so that true conclusions can be reached

PROBLEMS of INDUCTION



- **David Hume (1711–1776)** – made a distinction between systems of ideas that can provide certainty – e.g. math – and those that rely on our perceptions (empirical evidence) which are not certain.
- He recognized the importance of inductive thinking in the advancement of scientific knowledge but **highlighted its restrictions in finding the truth.**
- He set out in detail that inductive reasoning is not logically valid.

DEDUCTIVE REASONING

- **Deductive reasoning** begins with general statements (**premises**) and, through **logical argument**, comes to a specific **conclusion**.
- **Deduction**, as with many philosophical ideas, was first discussed as a way of reasoning by the Ancient Greeks, in particular, Plato.
- Enquiry is guided by the **theory** which precedes it.

All living things will eventually die. (General statement – first premise)

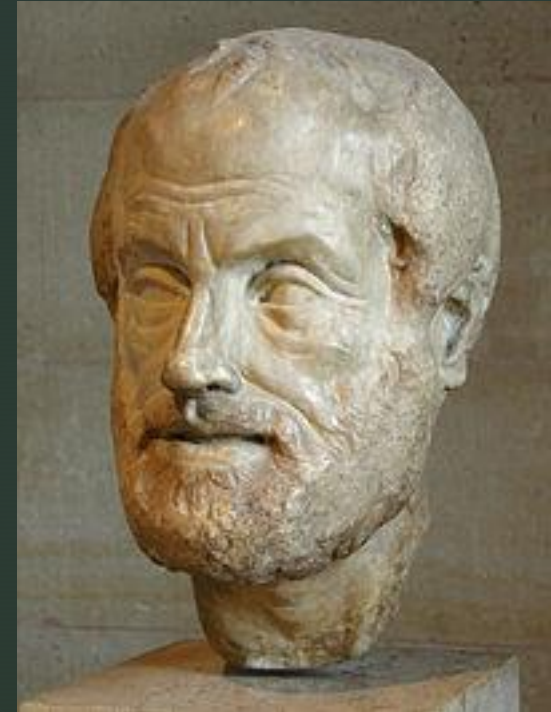
This animal is a living thing. (Inference – second premise)

Therefore, this animal will eventually die (Conclusion)

ARISTOTLE'S LOGIC

“A deduction is speech (logos) in which, certain things having been supposed, something different from those supposed results of necessity because of their being so.” (Prior Analytics I.2, 24b18–20)

- All Aristotle's logic revolves around one notion: the **deduction** (*sullogismos*).
- Each of the “things supposed” is a **premise** (*protasis*) of the argument, and what “results of necessity” is the **conclusion** (*sumperasma*).



PROBLEMS of DEDUCTION

- The validity of the argument does not guarantee the truth of the conclusion, the premises may be false.
- Deduction can never allow us to say anything new or informative about the world because the conclusion is already contained in the premises.



Burgess, 2007

WHAT IS THEORY?



“A scientific theory is a universal statement ”
(Popper, 1935)

A theory is: “a set of constructs, definitions and propositions that present a systematic view of phenomenon by specifying relations among variables” (Kerlinger, 1973)

“The purpose of a theory is to describe, explain and predict” (Borgatta & Montgomery, 2000)

WHAT IS THEORY?

- Theories are speculative answers to perceived problems and are **tested by observation and experiment**.
- Whilst it is possible to confirm the possible truth of a theory through observations which support it, theory can be falsified and totally rejected by making observations which are inconsistent with its statement.
- In this way, **science** is seen to proceed by trial and error: when one theory is rejected, another is proposed and tested, and thus the fittest theory survives.

TESTING THEORY

- In order for a theory to be tested, it must be expressed as a statement called a **hypothesis**.
- The essential nature of a hypothesis is that it must be **falsifiable**.
- This means that it must be logically possible to make true observational statements which conflict with the hypothesis, and thus can falsify it.
- The process of **falsification** leads to a devastating result of total rejection of a theory, requiring a completely new start.

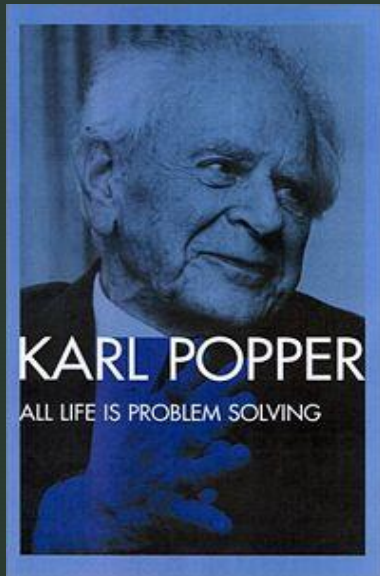
Nicholas Walliman, *Research Methods: The Basics* (2010)

HYPOTHETICO-DEDUCTIVE REASONING

- The hypothetico-deductive method combines inductive and deductive reasoning and overcomes the shortcomings of each to **acquire knowledge** (epistemology).
- It is this combination of experience with deductive and inductive reasoning which is **the foundation of modern scientific research** and is commonly referred to as **scientific method**.
- It was only by the beginning of the 1960s that Karl Popper (1902–1994) formulated the idea of the hypothetico-deductive method, even though it must have been used in practice for decades before.

Sir KARL POPPER (1902-1994)

“In so far as a scientific statement speaks about reality, it must be falsifiable; and in so far as it is not falsifiable, it does not speak about reality.”



Sir Karl Raimund Popper CH FBA FRS was an Austrian-born British philosopher and professor. Generally regarded as one of the 20th century's greatest philosophers of science, Popper is known for his rejection of the classical inductivist views on the scientific method in favour of **empirical falsification**. ([Wikipedia](#))



The to-and-fro process of:

- identification or clarification of a problem;
- developing a **hypothesis** (testable theory) **inductively** from **observations**;
- charting their implications by **deduction**;
- practical or theoretical **testing** of the hypothesis;
- rejecting or refining it in the light of the results.

SCIENTIFIC METHOD in Dictionaries

“The **principles and empirical processes of discovery and demonstration** considered characteristic of or necessary for scientific investigation, generally involving the **observation of phenomena**, the **formulation of a hypothesis** concerning the phenomena, **experimentation** to demonstrate the truth or falseness of the hypothesis, and a **conclusion** that validates or modifies the hypothesis.” (*The American Heritage Dictionary of the English Language, 4th Edition, 2004.*)

“An **orderly technique of investigation** that is supposed to account for scientific progress. The method consists of the following steps: (1) careful **observations** of nature. (2) **Deduction** of natural laws. (3) Formulation of **hypotheses** - generalization of those laws to previously unobserved phenomena. (4) Experimentation or **observational testing** of the validity of the predictions thus made. Actually, scientific discoveries rarely occur in this idealized, wholly rational, and orderly fashion.” (*The New Dictionary of Cultural Literacy, 3rd Edition, E.D. Hirsch, Jr., J.F. Kett, and J. Trefil, editors 2002.*)

SCIENTIFIC METHOD in Wikipedia

“A scientific method or process is considered fundamental to the scientific investigation and acquisition of new knowledge based upon **physical evidence**.

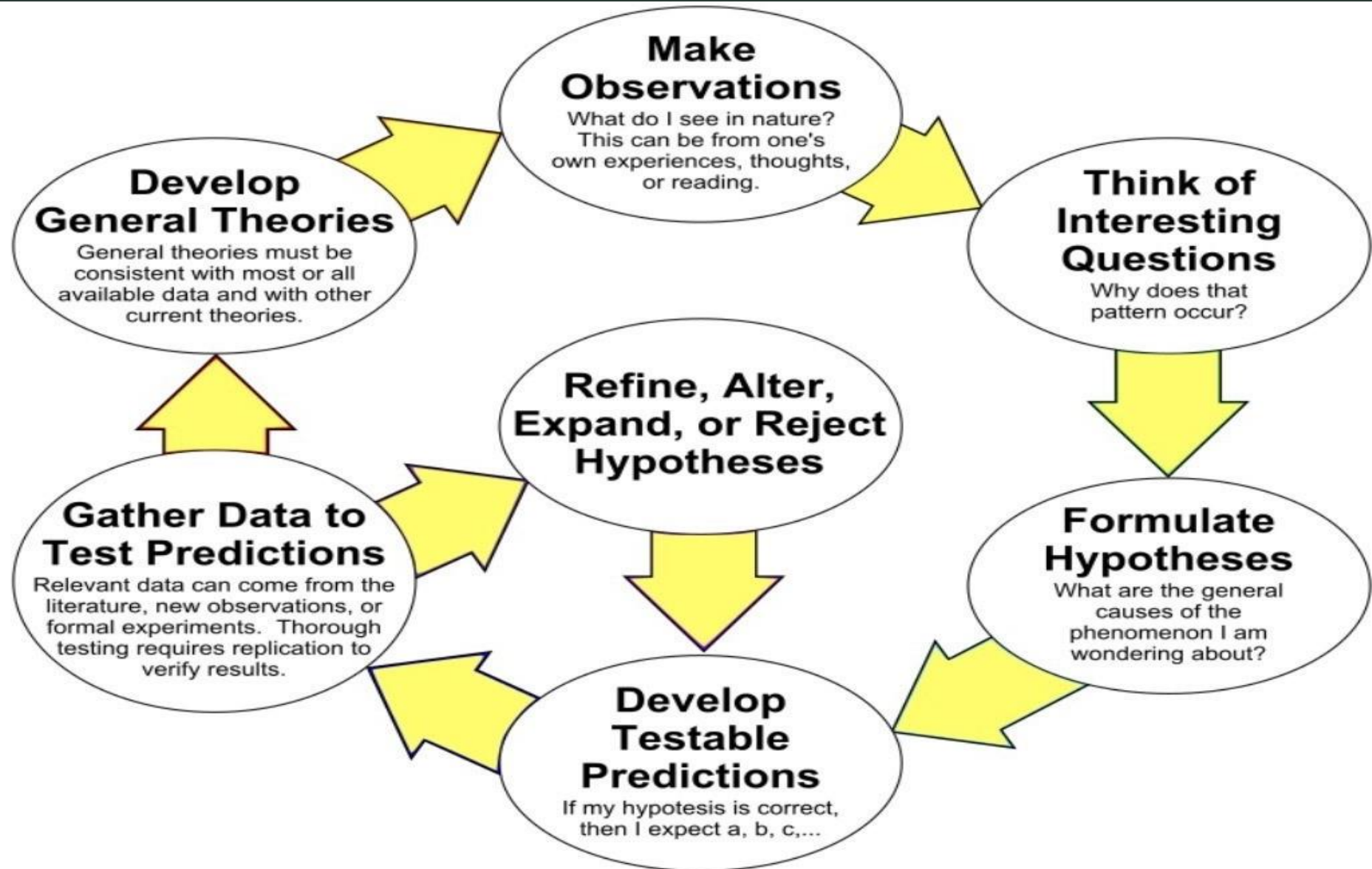
Scientists use **observation**, **hypotheses** and **deductions** to propose **explanations** for natural phenomena in the form of **theories**.

Predictions from these theories are tested by **experiment**. If a prediction turns out to be correct, the theory survives. Any theory which is cogent enough to make predictions can then be tested reproducibly in this way.

The method is commonly taken as the ***underlying logic of scientific practice***.

A scientific method is essentially an extremely cautious means of building a supportable, evidence-based understanding of our natural world.”

SCIENTIFIC METHOD as an Ongoing Process



PHILOSOPHY underlies SCIENTIFIC METHOD



There are certain assumptions that underlie scientific method that relate to a materialist view of metaphysics and a positivist view of epistemology.

These assumptions are:

- **Order** – the universe is an ordered system that can be investigated and the underlying ‘rules’ can be exposed.
- **External reality** – we all share the same reality that does not depend on our existence. We can therefore all equally contribute to and share knowledge that reflects this reality
- **Reliability** – we can rely on our senses and reasoning to produce facts that reliably interpret reality
- **Parsimony** – the simpler the explanation the better. Theories should be refined to the most compact formulation.
- **Generality** – the ‘rules’ of reality discovered through research can be applied in all relevant situations regardless of time and place.

Nicholas Walliman, *Research Methods: The Basics* (2010)

MATHEMATICAL vs. SCIENTIFIC METHOD

	<u>Mathematical method</u>	<u>Scientific method</u>
1	Understanding	Characterization from experience & observation
2	Analysis	Hypothesis: a proposed explanation
3	Synthesis	Deduction: prediction from the hypothesis
4	Review/Extend	Test and experiment



George Pólya's (1887 - 1985) work on problem solving, the construction of mathematical proofs, and heuristic show that mathematical method and scientific method differ in detail, while resembling each other in the use of iterative or recursive steps.

WHAT IS SCIENCE?

Conant, J.B. (*On Understanding Science: An Historical Approach*. New Haven, CT: Yale University Press, 1951) defined science as a connected series of theories and concepts derived from observation and experimentation that can lead to further experimentation and observation.

Science, therefore, can be viewed as the process of searching for explanations, or for the causes of events, and it is defined by its method: the **scientific method**

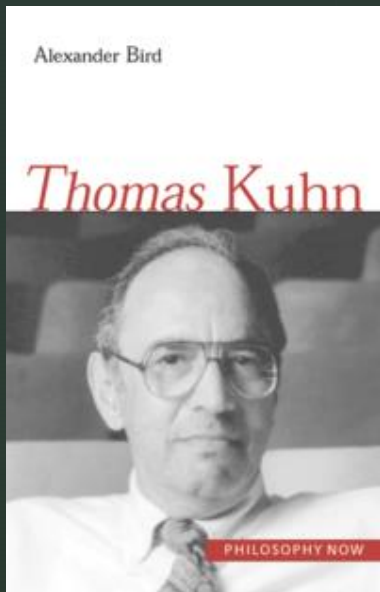


James Bryant Conant (1893 - 1978) was an American chemist, a transformative President of Harvard University, and the first U.S. Ambassador to West Germany. Conant obtained a PhD in Chemistry from Harvard in 1916. During World War I he served in the U.S. Army, working on the development of poison gases. ([Wikipedia](#))

WHAT IS SCIENCE?

“Science is a social activity, which is affected by practical activity, empirical observations and broad theoretical paradigms”

Thomas Kuhn (1922-1996)



The Structure of Scientific Revolutions (SSR) was originally printed as an article in the *International Encyclopedia of Unified Science*, published by the logical positivists of the Vienna Circle. In this book, Kuhn argued that science does not progress via a linear accumulation of new knowledge, but undergoes periodic revolutions, also called "paradigm shifts" (although he did not coin the phrase), in which the nature of scientific inquiry within a particular field is abruptly transformed. In general, science is broken up into three distinct stages. Prescience, which lacks a central paradigm, comes first. This is followed by "normal science", when scientists attempt to enlarge the central paradigm by "puzzle-solving". Guided by the paradigm, normal science is extremely productive: "when the paradigm is successful, the profession will have solved problems that its members could scarcely have imagined and would never have undertaken without commitment to the paradigm".

WHAT IS SCIENCE?



“The intellectual and practical activity encompassing the systematic study of the structure and behaviour of the physical and natural world through observation and experiment.”

Oxford English Dictionary

WHAT IS SCIENCE?

“Science is a **process** for:

- systematically **collecting** and recording **data** about the physical world,
- then categorizing and **studying** that data
- **to infer** the **principles** of nature that best explain the observed phenomena.”

Definition agreed by 72 Nobel laureates in 1986 according to
www.answersinscience.org/What-Is-Science.htm

GOALS of SCIENCE?

“*The ultimate goal of science is the development of a theory to explain the lawful relationships that exist in a particular field.*”



E. Bright Wilson, Jr.

*Theodore William Richards Professor of Chemistry,
Emeritus, Harvard University*

GOALS of SCIENCE?

- Description of Phenomena.
- Discovering Laws. **Law** - a statement that certain events are regularly associated with each other in an orderly way.
- Search for Causes.
 - We often overlook the real cause,
 - Some events are just coincidences,
 - Sometimes the real cause is another event correlated with the suspected cause,
 - Cause cannot happen after their effects.